LATENT WATER: COMPARING AND TRAVERSING NEURAL AUDIO LATENT SPACES

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ABSTRACT

This project compares the latent spaces of two neural audio models: EnCodec, an audio compression model based on Residual Vector Quantization, and RAVE, a real-time audio waveform synthesis model based on a Variational Autoencoder. Using three water-themed pieces by Maurice Ravel, Luciano Berio, and Billie Eilish, we extract and compare their latent representations through PCA, CKA, and MMD, finding that the two models capture fundamentally different aspects of musical audio. Building on this, we propose an artistic extension for an audiovisual installation, using barycentric blending, temporal reordering, and cross-space resynthesis to generate soundscapes that exist outside the boundaries of any single source piece.
Keywords: latent space, EnCodec, RAVE, audio representation, cross-space resynthesis, audiovisual installation

1. INTRODUCTION

The relationship between sound and its mathematical representation has been a central concern in both music technology and media art. Following Bourriaud's conception of art as the production of relations between people and the world [1] and the Sound Model Factory framework proposed by [2], this project, titled *Latent Water*, compares two latent spaces and treats them not merely as compression artifacts, but as a navigable creative space: a territory where sounds can be explored, manipulated, and transformed.

The two models used represent distinct approaches to this problem. EnCodec [3] is an audio compression model that prioritizes compression fidelity through a discrete token-based representation using Residual Vector Quantization (RVQ). RAVE [4], developed at IRCAM, is a real-time audio synthesis model that prioritizes smooth and controllable synthesis through a continuous VAE latent space, making it particularly suited for real-time performance and interactive applications. Despite operating on the same input domain, the two models encode fundamentally different aspects of audio, and the nature of this difference remains underexplored.

This project takes its conceptual inspiration from the tradition of audiovisual art, particularly the work of Nam June Paik, who pioneered the use of electronic media as a fluid, transformable material. Just as Paik treated the video signal as something to be distorted and reimagined, we treat the audio latent space as a malleable medium that can be stretched, interpolated, and traversed.

The three water-themed pieces used, *Jeux d'eau* (Ravel, 1901), *Wasserklavier* (Berio, 1965), and *Ocean Eyes* (Billie Eilish & blackbear, 2016), span three different musical periods, styles, and production contexts, making them well-suited for comparative latent space analysis. By encoding these pieces into both models, we ask: how do different neural representations hear the same sonic material, and what happens when we move between them?

The motivation behind this project is both academic and personal. The primary goal was to visualise and compare latent spaces, understanding how different models represent audio data. This analytical work is complemented by an artistic dimension, since the pipeline of this project will be used for an audiovisual installation currently in development for an academic conference for which I am leading the sound design.

2. METHODOLOGY

2.1 Dataset

Three water-themed pieces were selected (audio001–003, see repository). All audio outputs referenced in this paper are available at: https://github.com/polyxenipouliou/latent_water. A 30-second segment was extracted from the middle of each piece. Segments were resampled to 24kHz for EnCodec and 44.1kHz for RAVE (audio004–009, see repository).

2.2 EnCodec

EnCodec [3] was trained on a diverse dataset including music, speech, and environmental sounds. The model consists of an encoder E, a Residual Vector Quantization layer Q, and a decoder G that reconstructs the time-domain signal. We used $n_q = 8$ quantizers at 6kbps bandwidth. The encoder produces 75 latent frames per second, each described by a 128-dimensional vector, resulting in representations of shape $(2250, 128)$ for each 30-second segment (the decoded samples from this model are audio010,audio012,audio014 in the repository).

2.3 RAVE

We used the pretrained vintage RAVE model [4], trained on a dataset of vintage instruments at 44100Hz. Unlike EnCodec, RAVE employs a Variational Autoencoder

(VAE) architecture with adversarial fine-tuning, producing a continuous 16-dimensional latent space. The latent space is partitioned into informative and uninformative dimensions, facilitating smoother exploration of latent trajectories. For each 30-second segment, the model produced 646 frames, resulting in representations of shape $(646, 16)$ (the decoded samples from this model are audio011,audio013,audio015 in the repository).

2.4 Latent Space Comparison

To compare the two latent spaces, we used three complementary methods. Principal Component Analysis (PCA) was applied to reduce both representations to 2D for visualization. Centered Kernel Alignment (CKA) [5] was used as a similarity index between the two spaces, having been shown to outperform previous methods such as SVCCA [6]. Maximum Mean Discrepancy (MMD) with an RBF kernel was applied in three configurations: between pieces within the EnCodec space, between pieces within the RAVE space, and between the two spaces for the same piece, using the first 16 dimensions of EnCodec to match RAVE's dimensionality.

2.5 Artistic Extension Methods

Three latent space operations were developed. First, *temporal reordering*: each segment was divided into 15 chunks of approximately two seconds, reordered using a greedy nearest-neighbor algorithm by Euclidean distance in the latent space. Second, *barycentric blending*: latent vectors from all three pieces were mixed using weights sampled from a Dirichlet distribution, then reordered with the same NN algorithm. Third, *cross-space resynthesis*: the chunk ordering derived from the EnCodec latent space of one piece was applied to the RAVE latents of a different piece, which was then decoded by the RAVE model.

3. RESULTS

3.1 Statistical Descriptors

The EnCodec encoder produced latent representations of shape $(2250, 128)$ for each 30-second segment, corresponding to 75 frames per second. The RAVE encoder produced representations of shape $(646, 16)$, reflecting its lower temporal resolution and more compact latent dimensionality. The statistical descriptors of both spaces reveal important structural differences (Tables 1 and 2).

Piece	Min	Max	Mean
Ravel – *Jeux d'eau*	−19.757	24.255	−0.542
Berio – *Wasserklavier*	−15.422	21.031	−0.542
Billie Eilish – *Ocean Eyes*	−25.150	36.546	−0.557

Table 1. EnCodec latent space statistical descriptors.

In the EnCodec space, all pieces show a mean close to zero, consistent with a centered latent space. The dynamic range varies considerably: Berio has the narrowest range, followed by Ravel, while Billie Eilish shows the widest, suggesting that EnCodec captures more extreme acoustic features for contemporary pop production.

Piece	Min	Max	Mean
Ravel – *Jeux d'eau*	−3.758	3.748	0.001
Berio – *Wasserklavier*	−3.861	3.852	−0.031
Billie Eilish – *Ocean Eyes*	−3.851	4.994	0.172

Table 2. RAVE latent space statistical descriptors.

In the RAVE space, all pieces have a mean very close to zero, as expected from a VAE with a standard normal prior. The dynamic range is notably narrower (approximately $[-4, +4]$) compared to EnCodec ($[-25, +36]$), reflecting the VAE's regularization. Billie Eilish is the only piece showing a slightly wider range and a non-negligible mean (0.172), suggesting it occupies a more peripheral region of RAVE's learned distribution.

3.2 PCA Visualisation

The PCA plots reveal a key structural difference between the two models (Figures 1 and 2). In the EnCodec space, Ravel and Berio cluster tightly together while Billie Eilish extends far along PC1. In the RAVE space, Ravel and Berio are more clearly separated, indicating that the vintage-trained model captures finer timbral distinctions. Billie Eilish remains an outlier in both spaces.

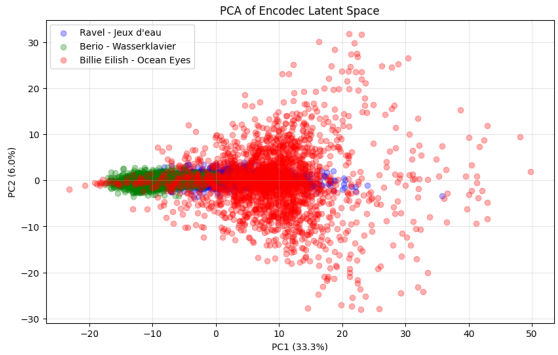

Figure 1. PCA plot of EnCodec Latent Space.

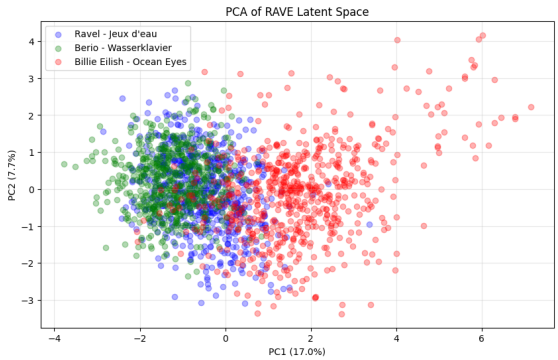

Figure 2. PCA plot of RAVE Latent Space.

3.3 Centered Kernel Alignment (CKA)

One of the challenges in interpreting neural networks is their black-box behaviour. [5] investigated the similarity between neural network representations using CKA as a similarity index, which has been shown to outperform previous methods, such as SVCCA in [6]. The CKA analysis reveals very low similarity between the two latent spaces across all three pieces (Table 3), confirming that EnCodec and RAVE encode fundamentally different aspects of the audio. Ravel's *Jeux d'eau* shows the highest similarity (0.059), likely because its acoustic piano timbre is closest to RAVE's vintage training domain. Billie Eilish shows the lowest (0.024), reflecting the greater distance between contemporary pop production and the vintage corpus.

Piece	CKA (EnCodec, RAVE)
Ravel – *Jeux d'eau*	0.059
Berio – *Wasserklavier*	0.026
Billie Eilish – *Ocean Eyes*	0.024

Table 3. CKA similarity between EnCodec and RAVE latent spaces.

3.4 Maximum Mean Discrepancy (MMD)

While CKA measures the similarity between two representational spaces as a whole, MMD quantifies the distance between two probability distributions using a Radial Basis Function (RBF) kernel in a reproducing kernel Hilbert space (RKHS). A value close to zero indicates similar distributions; larger values indicate greater divergence.

Pair	EnCodec	RAVE
Ravel vs Berio	0.0040	0.0042
Ravel vs Billie Eilish	0.0040	0.0044
Berio vs Billie Eilish	0.0040	0.0047

Table 4. MMD between pieces within each latent space.

Piece	MMD (EnCodec[:16] vs RAVE)
Ravel – *Jeux d'eau*	0.0044
Berio – *Wasserklavier*	0.0091
Billie Eilish – *Ocean Eyes*	0.0044

Table 5. Cross-model MMD per piece.

In the EnCodec space, all pairwise distances are identical (0.0040), indicating limited discriminative power (Table 4). In the RAVE space, a consistent gradient emerges: lower distances between the two piano works and higher distances between Berio and Billie Eilish. Cross-model comparisons (Table 5) show that Berio yields the highest divergence (0.0091). A possible explanation is that Berio's extended techniques and sparse textures occupy a region of the EnCodec space with no close correspondence in RAVE's distribution, whereas Ravel's conventional writing and Billie Eilish's spectrally consistent production both map more smoothly onto RAVE's space, albeit for different reasons.

4. ARTISTIC EXTENSION

The goal of this section is to use EnCodec's latent space to create disorienting soundscapes for an audiovisual installation on the theme of washing. Following Bourriaud's notion of the artwork as a social interstice [1], the resulting audio accompanies a looping installation where the cyclical quality of the reordering process mirrors the repetitive gestures of washing.

4.1 Temporal Reordering via Latent Similarity

Each audio segment is divided into 15 equal chunks of approximately two seconds. The mean latent vector of each chunk is computed, and chunks are reordered using a greedy nearest-neighbor algorithm: each subsequent chunk is the nearest unvisited one by Euclidean distance. The result is a new temporal sequence with a completely reorganized macrostructure (audio016, see repository).

4.2 Barycentric Blending and Latent Reordering

Latent vectors from all three pieces are blended using weights sampled from a Dirichlet distribution, ensuring they always sum to 1.0. The blended signal is divided into chunks and reordered with the same NN algorithm. The result is an audio that does not belong to any of the three originals, with a temporal structure dictated by the geometry of the blended latent space. audio017 (w_{Ravel}=0.31, w_{Berio}=0.45, w_{Billie}=0.25) and audio018 (w_{Ravel}=0.02, w_{Berio}=0.60, w_{Billie}=0.38) are available in the repository.

4.3 Cross-Space Resynthesis

The chunk ordering is derived from the EnCodec latent space of one piece and applied to the RAVE latents of a randomly selected different piece, which is then decoded by the RAVE model. The timbral identity is shaped by RAVE's vintage-trained decoder, while the temporal macrostructure follows the latent geometry of a different piece entirely. This demonstrates the compositional possibilities that emerge from treating the two latent spaces as independent, interchangeable axes of sonic organisation (audio019–022, see repository).

5. CONCLUSION AND FUTURE WORK

This project compared the latent spaces of EnCodec and RAVE, finding that the two models encode fundamentally different aspects of audio. EnCodec produces a wide discrete space with limited sensitivity to stylistic variation, while RAVE's continuous VAE space captures finer timbral distinctions — though this is also related to the specific pretrained model used. CKA and MMD results confirmed low correspondence between the two representations across all three pieces.

These findings informed the artistic extension: by treating the two spaces as independent axes of sonic organisation, the cross-space resynthesis operations produced soundscapes that belong to neither source, intended for a looping installation on the theme of washing. Future work could explore learned mappings between the two latent spaces and extend the pipeline to real-time synthesis for live performance contexts.

6. REFERENCES

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